1 Continuous Functions in Several Variables

Definition 1.1 (Continuity)**.** For $X \subseteq \mathbb{R}^n$, $Y \subseteq \mathbb{R}^p$ and $f : X \to Y$, f is *continuous at* $a \in X$ if for every $\varepsilon > 0$ there is $\delta > 0$ with

$$|f(x) - f(a)| < \varepsilon \quad \text{for all } x \in X \text{ with } |x - a| < \delta,$$

where $|x| = \sqrt{\sum_{j=1}^{n} x_j^2}$ is the Euclidean norm. Everything that survived the step from $\mathbb{R}$ to $\mathbb{C} = \mathbb{R}^2$ (sequences, series, limit rules) survives the step to $\mathbb{R}^n$.

Example 1.1. $f(x, y) = \frac{xy}{x^2+y^2}$ (and 0 at the origin) is discontinuous at $(0, 0)$ although every section $x \mapsto f(x, b)$, $y \mapsto f(a, y)$ is continuous: separate continuity does not imply continuity.

Definition 1.2 (Balls, spheres, open and closed sets)**.** $U_\varepsilon(a) = \{x \mid |x - a| < \varepsilon\}$ (open ball), $D_\varepsilon(a)$ (closed ball), $S_\varepsilon(a)$ (sphere). $U \subseteq \mathbb{R}^n$ is *open* if every $a \in U$ has some $U_\varepsilon(a) \subseteq U$; F is *closed* if $\mathbb{R}^n \setminus F$ is open. $U \subseteq X$ is *relatively open in* X if $U = V \cap X$ for an open V; dually for relatively closed.

Proposition 1.1. *$\emptyset$ and X are relatively open and closed in X; finite intersections and arbitrary unions of (relatively) open sets are open; dually for closed sets.*

Theorem 1.1 (Global description of continuity)**.** *For $f : X \to Y$ equivalent:*

- *f is continuous;*
- *$f^{-1}(V)$ is open in X for every (relatively) open $V \subseteq Y$;*
- *$f^{-1}(G)$ is closed in X for every closed $G \subseteq Y$.*

Example 1.2. Closed balls and spheres are closed ($D_\varepsilon(a) = f^{-1}[0, \varepsilon]$ for $f(x) = |x - a|$); affine subspaces are closed (fibres of continuous linear maps); intervals are the open (resp. closed) subsets of $\mathbb{R}$ in the old sense.

Lemma 1.1 (Sequential description)**.** *X is closed iff every sequence in X that converges in $\mathbb{R}^n$ has its limit in X.*

Definition 1.3 (Compactness)**.** $X \subseteq \mathbb{R}^n$ is *compact* if every sequence in X contains a subsequence that converges in X (to a point of X).

Theorem 1.2 (Continuous images of compact sets)**.** *If X is compact and $f : X \to \mathbb{R}^p$ continuous, then $f(X)$ is compact.*

Sketch. *Lift a sequence in $f(X)$ to X, extract a convergent subsequence there, and use continuity.*

Theorem 1.3 (Heine–Borel)**.** *$X \subseteq \mathbb{R}^n$ is compact iff it is closed and bounded.*

Sketch. *Compact ⇒ closed: limits of convergent sequences lie in X; bounded: else $|x_j| > j$ has no convergent subsequence. Conversely, bounded sequences have convergent subsequences (Bolzano–Weierstrass), and closedness brings the limit into X.*

Lemma 1.2. *Every nonempty compact subset of $\mathbb{R}$ has a smallest and a largest element ($\sup X, \inf X \in X$). Hence a continuous real function on a compact set attains a minimum and a maximum.*

Definition 1.4 (Path-connectedness)**.** $X \subseteq \mathbb{R}^n$ is *path-connected* if any $a, b \in X$ can be joined by a continuous path $\gamma : [0, 1] \to X$ with $\gamma(0) = a$, $\gamma(1) = b$. Balls (open or closed) and spheres of positive dimension are path-connected.

Lemma 1.3. *Continuous images of path-connected sets are path-connected; the path-connected subsets of $\mathbb{R}$ are exactly the intervals.*

Example 1.3. Every quadratic form $q : \mathbb{R}^n \to \mathbb{R}$ is continuous, so $q(S^{n-1})$ is a compact interval $[c, d]$: q attains on the unit sphere a minimum c and a maximum d and everything in between. Since $q(x) = r^2 q(x/r)$ on rays, q is positive definite iff $c > 0$; by the principal axis theorem c and d are the smallest and largest eigenvalues of q.

Theorem 1.4 (Inverse maps)**.** *If X is compact and $f : X \to Y$ continuous and bijective, then $f^{-1} : Y \to X$ is continuous.*

2 Null Sets & the One-Dimensional Integral

Definition 2.1 (Null set)**.** $X \subseteq \mathbb{R}^n$ is a *null set* if for every $\varepsilon > 0$ there is a sequence of rectangles (boxes) (Q_j) with

$$X \subseteq \bigcup_{j=0}^{\infty} Q_j \quad \text{and} \quad \sum_{j=0}^{\infty} \mu(Q_j) < \varepsilon.$$

Lemma 2.1. *Every subset of a null set is a null set; countable unions of null sets are null sets.*

Example 2.1. For $n > 0$ every countable subset of $\mathbb{R}^n$ is a null set (so $\mathbb{Q}$ is one); the coordinate hyperplane $\mathbb{R}^{n-1} \times \{0\}$ and, more generally, every proper affine subspace of $\mathbb{R}^n$ is a null set.

Definition 2.2 (Extension by zero, integrability)**.** For $f : X \to \mathbb{R}$ let $f_X : \mathbb{R}^n \to \mathbb{R}$ be f on X and 0 elsewhere. The n-dimensional integral is a linear functional $f \mapsto \int f$ on a vector space $L^1(\mathbb{R}^n)$ of *integrable* functions, with:

- if $f \in L^1$ then $f \cdot \mathbf{1}_{\{g>0\}} \in L^1$ for $g \in L^1$;
- $f \geq 0 \Rightarrow \int f \geq 0$ (monotonicity in particular);
- if X is compact and $f : X \to \mathbb{R}$ continuous then $f_X \in L^1$;
- for a compact box Q: $\int \mathbf{1}_Q = \mu(Q)$ (its volume);
- if X is a null set, then $f_X \in L^1$ and $\int f_X = 0$ for *every* $f : X \to \mathbb{R}$.

Statements hold "almost everywhere" (a.e.) if they fail only on a null set.

Proposition 2.1 (Triangle inequality, positive/negative part)**.** *With $f = f^+ - f^-$ and $|f| = f^+ + f^-$ (where $f^+ = \max(f, 0) \geq 0$): for $f \in L^1(\mathbb{R}^n)$ also $|f| \in L^1$ and*

$$\left|\int f\right| \leq \int |f|.$$

Theorem 2.1 (Mean value theorem for integrals)**.** *If X is compact and path-connected and $f : X \to \mathbb{R}$ continuous, there is $t \in X$ with*

$$\int_X f = f(t) \cdot \int_X 1 = f(t) \cdot \mu(X),$$

where $\mu(X) = \int_X 1$ is the volume (measure) of X.

Sketch. *$f(X)$ is a compact interval $[c, d]$; the quotient $\int_X f/\mu(X)$ lies in it.*

Theorem 2.2 (Fundamental theorem of calculus)**.** *If $f : I \to \mathbb{R}$ is continuous and $a \in I$, then $x \mapsto \int_a^x f$ is an antiderivative: $\frac{\mathrm{d}}{\mathrm{d}x} \int_a^x f = f(x)$. Consequently, for an antiderivative F of the continuous f on $[a, b]$:*

$$\int_a^b f = F(b) - F(a) \qquad \textit{(Newton–Leibniz).}$$

Proposition 2.2 (Integration rules on $[a, b]$)**.** •
Additivity: $\int_{(a,c)} f = \int_{(a,b)} f + \int_{(b,c)} f$ (up to null sets).

- *Partial integration:* $\int_a^b f'g = (fg)\big|_a^b - \int_a^b fg'$ *for* $f, g \in C^1$.
- *Substitution:* $\int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,\mathrm{d}x = \int_\alpha^\beta f(\varphi(t))\,\varphi'(t)\,\mathrm{d}t$ *for* $\varphi \in C^1$; φ *need not be monotone.*

Example 2.2. Linear substitution; $\int_0^{2\pi} f(\cos t)\sin t\,\mathrm{d}t = 0$ by substituting $x = \cos t$. A rational function has locally $\log|x-c|$ as an antiderivative of $1/(x-c)$; for complex poles combine conjugate terms to get real arctan and log expressions.

3 Integral and Limit

Definition 3.1 (Integral seminorm)**.** $\|f\| := \int|f| \in [0,\infty]$ for $f \in L^1(\mathbb{R}^n)$. Properties: $\|f\| \geq 0$, $\|f\| = 0 \iff f = 0$ a.e., $\|\alpha f\| = |\alpha|\|f\|$, $\|f \pm g\| \leq \|f\| + \|g\|$.

Remark 3.1. Integral and limit are *not* generally interchangeable: $f_j = \mathbf{1}_{[j,j+1]}$ converges to 0 pointwise but $\int f_j = 1$ for all j. The following theorems give the conditions under which exchange is allowed.

Theorem 3.1 (Convergence axiom)**.** *For every Cauchy sequence* (f_j) *in* $L^1(\mathbb{R}^n)$*: some subsequence converges pointwise a.e.; all a.e. limits agree a.e. and are integrable; and* $\lim_j \|f_j - f\| = 0$*, i.e.* $\lim_j \int f_j = \int f$.

Theorem 3.2 (Monotone convergence)**.** *If* $f_0 \leq f_1 \leq \cdots$ *a.e. in* L^1 *and* $(\int f_j)$ *is bounded, then* (f_j) *converges a.e. to some* $f \in L^1$*, and* $\lim_j \int f_j = \int f$.

Theorem 3.3 (Dominated convergence)**.** *If* $f_j \to f$ *a.e. pointwise, all* $f_j \in L^1$*, and* $|f_j| \leq g$ *a.e. for some* $g \in L^1$*, then* $f \in L^1$ *and*

$$\lim_j \int f_j = \int f.$$

Theorem 3.4 (Exhaustion)**.** *If* $X_0 \subseteq X_1 \subseteq \cdots$ *with* $\bigcup_j X_j = X$ *and* f *is integrable over every* X_j*, then* f *is integrable over* X *iff the increasing sequence* $\int_{X_j}|f|$ *is bounded; then* $\lim_j \int_{X_j} f = \int_X f$.

Theorem 3.5 (Improper integrals)**.** *For open (possibly unbounded)* (a,b) *and continuous* f*:* f *is integrable over* (a,b) *iff* $\lim_{\substack{a'\to a\\ b'\to b}} \int_{a'}^{b'}|f|$ *exists; then* $\int_{(a,b)} f$ *equals that limit of* $\int f$.

Example 3.1. $\int_0^1 x^{-1/2}\,\mathrm{d}x = 2$, but $x \mapsto x$ is not integrable over $\mathbb{R}$ even though $\int_{-r}^r x\,\mathrm{d}x = 0$ (absolute integrability fails). The Gamma function $\Gamma(x) = \int_0^\infty t^{x-1}\mathrm{e}^{-t}\,\mathrm{d}t$ is finite for $x > 0$ and satisfies $\Gamma(x+1) = x\Gamma(x)$, $\Gamma(k+1) = k!$.

Theorem 3.6 (Parameter-dependent integrals)**.** *Let* X *be open* $\cap$ *closed (e.g. open or closed),* $T \subseteq \mathbb{R}^p$ *compact,* $f : X \times T \to \mathbb{R}$ *continuous. Then*

$$F : X \to \mathbb{R}, \quad F(x) := \int_T f(x,t)\,\mathrm{d}t$$

is continuous. If moreover $\frac{\partial f}{\partial x_j}$ *exists and is continuous, then* F *is differentiable with*

$$\frac{\partial F}{\partial x_j}(x) = \int_T \frac{\partial f}{\partial x_j}(x,t)\,\mathrm{d}t.$$

4 Measures, Fubini & Cavalieri

Definition 4.1 (Measurable sets)**.** $X \subseteq \mathbb{R}^n$ is *measurable* if $\mathbf{1}_{D_r(0)\cap X} \in L^1(\mathbb{R}^n)$ for every $r \in \mathbb{N}$; then

$$\mu(X) := \lim_{r\to\infty} \int \mathbf{1}_{D_r(0)\cap X} \in [0,\infty]$$

is the *measure* (volume) of X.

Theorem 4.1. *Open and closed sets are measurable; complements, countable unions and countable intersections of measurable sets are measurable; for pairwise disjoint measurable* (X_j)*:* $\mu(\bigcup_j X_j) = \sum_j \mu(X_j)$*. The null sets are exactly the measurable sets of measure zero. If* $f \in L^1(\mathbb{R}^n)$ *and* X *is measurable, then* $f_X \in L^1(\mathbb{R}^n)$.

Theorem 4.2 (Fubini)**.** *Split* $\mathbb{R}^n = \mathbb{R}^k \times \mathbb{R}^l$ *and write* $f_y(x) = f(x,y)$*. If* $f \in L^1(\mathbb{R}^n)$*, then for almost all* $y \in \mathbb{R}^l$ *the function* f_y *is integrable over* $\mathbb{R}^k$*, the function* $y \mapsto \int f_y$ *(arbitrarily completed on a null set) is integrable, and*

$$\int_{\mathbb{R}^n} f = \int_{\mathbb{R}^l} \left(\int_{\mathbb{R}^k} f_y\,\mathrm{d}x \right) \mathrm{d}y.$$

Example 4.1. For $f(x,y) = \frac{y^2-x^2}{(x^2+y^2)^2}$ on $(0,1)^2$ both iterated integrals exist but are *not* equal ($1/2$ vs. $-1/2$): f is not integrable, and Fubini must not be applied. Iterated integrals alone say nothing.

Proposition 4.1 (Fubini on subsets)**.** *If* $X = \bigcup_j X_j$ *is an increasing exhaustion and* f *is integrable over each* X_j *(e.g. continuous on a compact* X*), then integrability over* X *and the value of* $\int_X f$ *may be tested via iterated integrals over the slices* $X^y = \{x \mid (x,y) \in X\}$.

Theorem 4.3 (Cavalieri's principle)**.** *For measurable* $X \subseteq \mathbb{R}^k \times \mathbb{R}^l$*: the slices* X^y *are measurable for a.e.* y*, and* $\mu(X) < \infty$ *iff* $y \mapsto \mu_k(X^y)$ *is integrable; then*

$$\mu_n(X) = \int_{\mathbb{R}^l} \mu_k(X^y)\,\mathrm{d}y.$$

Example 4.2. For compact X and continuous $f \geq 0$ on X, the region under the graph, $Y = \{(x,y) \in X \times \mathbb{R} \mid 0 \leq y \leq f(x)\}$, has $\mu_{n+1}(Y) = \int_X f$.

5 Differentiation in Several Variables

Definition 5.1 (Differential)**.** Let $X \subseteq \mathbb{R}^n$ be open, $a \in X$. Then $f : X \to \mathbb{R}^p$ is *differentiable at* a if there is a linear map $l : \mathbb{R}^n \to \mathbb{R}^p$ with

$$f(x) = f(a) + l(x-a) + o(|x-a|) \quad (x \to a).$$

This l is then uniquely determined; it is called the *differential* $Df(a)$ of f at a.

Sketch. Uniqueness: if l, m both qualify, test along lines $x = a + ty$; $(l-m)(y) = o(1)$ for $t \to 0$ forces $l = m$.

Theorem 5.1 (Chain rule)**.** *If* $f : X \to Y$ *and* $g : Y \to \mathbb{R}^q$ *are differentiable, then* $g \circ f$ *is differentiable with*

$$D(g \circ f)(a) = Dg\big(f(a)\big) \circ Df(a)$$

— the differential of a composition is the composition of the differentials (matrix product).

Example 5.1 (Bilinear maps)**.** Every bilinear $f : \mathbb{R}^m \times \mathbb{R}^n \to \mathbb{R}^p$ is differentiable with

$$Df(a,b)(x,y) = f(x,b) + f(a,y),$$

since the quadratic remainder $f(x-a, y-b) = O(|(x,y) - (a,b)|^2)$.

Definition 5.2 (Partial derivatives, Jacobi matrix)**.** The *partial derivative* is $D_j f_i(a) = \frac{\partial f_i}{\partial x_j}(a)$ (differentiate f_i in the j-th coordinate only). The *Jacobi matrix* is $Df(a) = \left(\frac{\partial f_i}{\partial x_j}(a)\right) \in \mathrm{Mat}(p \times n, \mathbb{R})$.

Lemma 5.1. *If f is differentiable at a, all partial derivatives exist there and $Df(a)$ is the Jacobi matrix. The converse is false.*

Theorem 5.2 (C^1 criterion)**.** *If all partial derivatives of $f : X \to \mathbb{R}^p$ exist and are continuous on X, then f is differentiable (a C^1-map).*

Sketch. *Walk from a to x along axis-parallel segments and apply the one-dimensional mean value theorem on each.*

Example 5.2 (Spherical coordinates)**.** $(r, \theta, \varphi) \mapsto (r\sin\theta\cos\varphi,\ r\sin\theta\sin\varphi,\ r\cos\theta)$ is C^1 with the well-known 3×3 Jacobi matrix; its determinant is $-r^2\sin\theta$.

Definition 5.3 (Curves)**.** A continuous $\gamma : T \to Y$ ($T \subseteq \mathbb{R}$ an interval) is a *parametrized curve*; for differentiable γ the *velocity* is $\dot\gamma(t)$, the *acceleration* $\ddot\gamma(t)$. The *derivative of f along v* (directional derivative) is $Df(a)v$: realize v as the velocity of a curve through a and test:

$$(f \circ \gamma)'(t) = Df(\gamma(t)) \cdot \dot\gamma(t) \quad \text{if } \gamma(t) = a,\ \dot\gamma(t) = v.$$

Definition 5.4 (Gradient)**.** For scalar f, the *gradient* is defined by $Df(a)v = \langle \operatorname{grad} f(a), v\rangle$ for all v, i.e. $\operatorname{grad} f(a) = Df(a)^t$ (row turned into a column).

Lemma 5.2 (Complex differentiability)**.** *$f = u + \operatorname{Im} v$ on an open $X \subseteq \mathbb{C}$ is complex differentiable at c iff it is differentiable as a real map and the* Cauchy–Riemann equations *hold:*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \qquad \text{at } c.$$

6 Diffeomorphisms, Inverse & Implicit Functions

Definition 6.1 (Diffeomorphism)**.** A differentiable $f : X \to Y$ between open sets is a *diffeomorphism* if a differentiable $g : Y \to X$ with $g \circ f = \mathrm{id}$ and $f \circ g = \mathrm{id}$ exists (equivalently: f is bijective and f^{-1} differentiable). One-dimensional criterion: $f : I \to \mathbb{R}$ is a diffeomorphism onto its (open) image iff $f' \neq 0$ throughout.

Lemma 6.1. *If f is a diffeomorphism, then every $Df(a)$ is an isomorphism; in particular $n = p$ (if $X \neq \emptyset$), and*

$$D(f^{-1})\big(f(a)\big) = Df(a)^{-1}.$$

Sketch. *Differentiate $f^{-1} \circ f = \mathrm{id}$ with the chain rule.*

Theorem 6.1 (Inverse function theorem)**.** *$f : X \to \mathbb{R}^n$ (X open, $f \in C^1$) is a local C^1-diffeomorphism at a iff $Df(a) \in \mathrm{GL}(n, \mathbb{R})$: locally around a (and around $f(a)$) f has a differentiable inverse.*

Theorem 6.2 (Openness)**.** *A local diffeomorphism maps open sets to open sets; if it is injective, it is a (global) diffeomorphism onto its open image.*

Corollary 6.1. *Restricting a (local/global) diffeomorphism to an open subset again gives one of the same kind.*

Theorem 6.3 (Implicit function theorem)**.** *Let $F : X \times Y \to \mathbb{R}^p$ be C^1 ($X \subseteq \mathbb{R}^n$, $Y \subseteq \mathbb{R}^p$ open), $F(a, b) = 0$. If the partial Jacobi matrix $\frac{\mathrm{d}F}{\mathrm{d}y}(a, b) \in \mathrm{GL}(p, \mathbb{R})$, then around (a, b) the zero set*

$$\{(x, y) \in U \times V \mid F(x, y) = 0\}$$

is the graph of a C^1 map $f : U \to V$: the equation $F(x, y) = 0$ determines $y = f(x)$ differentiably.

Example 6.1. Simple zeros of a polynomial depend differentiably on the coefficients: apply the implicit function theorem to the general polynomial $F(x, y) = y^n + \sum_{j<n} x_j y^j$ at a simple root.

Example 6.2 (Polar coordinates)**.** $(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$ has the *Jacobi determinant* $\det D(r, \theta) = r$; it is a local diffeomorphism exactly off the line $r = 0$.

7 Submanifolds

Definition 7.1 (Cards)**.** A *(differentiable) card* for an open $X \subseteq \mathbb{R}^n$ is a diffeomorphism $h : U \to h(U)$ from an open $U \subseteq X$ onto an open $h(U) \subseteq \mathbb{R}^n$; a card with $h(a) = 0$ is a card *around* a. If (U, h), (V, k) are cards and $f(U) \subseteq V$, then "f in the cards h, k" means $k \circ f \circ h^{-1}$; e.g. a scalar function in polar coordinates is $f \circ h^{-1}$ with h the inverse of the polar map.

Definition 7.2 (Rank, regular and critical points)**.** For $f : X \to \mathbb{R}^p$ the *rank at a* is $\operatorname{rk} Df(a)$. The point a is *regular* if $\operatorname{rk} Df(a) = p$ (only possible for $n \geq p$), otherwise *critical*.

Theorem 7.1 (Regular point theorem)**.** *Around every regular point a of a C^1-map f there is a card in which f is affine-linear:*

$$(f \circ h^{-1})(x_1, \ldots, x_n) = f(a) + (x_1, \ldots, x_p)^t.$$

Definition 7.3 (Submanifold)**.** $Y \subseteq \mathbb{R}^n$ is a *k-dimensional submanifold* of $\mathbb{R}^n$ if around every $a \in Y$ there is a card h with

$$h(U \cap Y) = h(U) \cap (\mathbb{R}^k \times \{0\}).$$

The card "flattens" Y to $\mathbb{R}^k \times \{0\}$.

Definition 7.4 (Regular values)**.** $b \in \mathbb{R}^p$ is a *regular value* of $f : X \to \mathbb{R}^p$ if *every* $a \in f^{-1}\{b\}$ is a regular point (values not attained count as regular).

Theorem 7.2 (Regular value theorem)**.** *If b is a regular value of the C^1-map $f : X \to \mathbb{R}^p$, then the fibre $f^{-1}\{b\}$ is a submanifold of dimension $n - p$.*

Sketch. *In a regular-point card, $f \circ h^{-1}$ is the projection onto the last p coordinates; its zero set is visibly $\mathbb{R}^{n-p} \times \{0\}$.*

Example 7.1. The sphere $S^{n-1} = f^{-1}\{1\}$ for $f(x) = |x|^2$ is an $(n-1)$-dimensional submanifold; $O(n)$ is a submanifold of $\mathrm{Mat}(n \times n, \mathbb{R})$ (fibre of $x \mapsto x^t x$ at $\mathbf{1}$); configuration spaces of point particles with differentiable constraints (fixed distances etc.) are submanifolds of $(\mathbb{R}^3)^n$.

8 Tangent Vectors & Lagrange Multipliers

Definition 8.1 (Tangent space)**.** For an n-dimensional submanifold $X \subseteq \mathbb{R}^N$ and $a \in X$, with a submanifold card H around a:

$$T_a X \coloneqq DH(a)^{-1}\big(\mathbb{R}^n \times \{0\}\big) \subseteq \mathbb{R}^N.$$

This subspace is independent of the card; its elements are *tangent vectors*.

Lemma 8.1 (Tangent vectors are velocities)**.** *$T_a X$ consists exactly of the velocity vectors $\dot\gamma(0)$ of differentiable curves γ in X with $\gamma(0) = a$.*

Definition 8.2 (Differential between manifolds)**.** $f : X \to Y$ between submanifolds is differentiable at a if in some submanifold card h for X around a the map $f \circ h^{-1}$ is differentiable at 0. The *differential* is

$$T_a f = D(f \circ h^{-1})(0) : T_a X \to T_{f(a)} Y,$$

independent of the card; the chain rule $T_a(g \circ f) = T_{f(a)} g \circ T_a f$ holds. If f extends to a differentiable F on an open neighborhood, then $T_a f = DF(a)\big|_{T_a X}$.

Theorem 8.1 (Tangent space of a fibre)**.** *If b is a regular value of the C^1-map $f : X \to \mathbb{R}^p$, then for every $a \in Y = f^{-1}\{b\}$:*

$$T_aY = \ker Df(a).$$

Sketch. *"$\subseteq$": $f|_Y$ is constant. Equality because both spaces have dimension $n-p$.*

Example 8.1. $T_aS^{n-1} = \{a\}^\perp$: the tangent space of the sphere is the hyperplane perpendicular to the position vector (since $Df(a) = 2a^t$ for $f(x) = |x|^2$).

Theorem 8.2 (Lagrange multipliers)**.** *Let b be a regular value of $F : X \to \mathbb{R}^p$ ($X \subseteq \mathbb{R}^n$ open), $Y = F^{-1}\{b\}$, and $f : X \to \mathbb{R}$ differentiable, $a \in Y$. Equivalent:*

- *a is a critical point of $f|_Y$ ("f under the constraint $F = b$");*
- *there are uniquely determined* Lagrange multipliers *$\lambda_1, \dots, \lambda_p \in \mathbb{R}$ such that a is a critical point of*

$$f - \lambda_1 F_1 - \cdots - \lambda_p F_p.$$

Example 8.2. For a quadratic form q with symmetric matrix s restricted to the sphere: critical points of $q - \lambda|x|^2$ solve $sx = \lambda x$; the critical points are the unit eigenvectors, the critical values the eigenvalues — the smallest and largest eigenvalues of q (cf. Section 1).

Proposition 8.1 (C^k maps, symmetry of mixed derivatives)**.** *$f : X \to \mathbb{R}^p$ is C^k iff all partial derivatives of order $\le k$ exist and are continuous. For C^k-maps the order of differentiation does not matter:*

$$D_{j_1} D_{j_2} \cdots D_{j_k} f = D_{j_{\sigma(1)}} \cdots D_{j_{\sigma(k)}} f \quad \text{for all } \sigma \in \mathrm{Sym}_k .$$

Sketch. *Case $k = 2$, $n = 2$: apply the one-dimensional mean value theorem twice in the two orders.*

Definition 8.3 (Tangent bundle, vector fields)**.** The *tangent bundle* of an n-dimensional C^k-submanifold is

$$TX \coloneqq \big\{(x, v) \in X \times \mathbb{R}^N \mid v \in T_xX\big\} \xrightarrow{\pi} X,$$

with fibres $\pi^{-1}\{x\} = \{x\} \times T_xX$. A *vector field* is a map $v : X \to TX$ with $\pi \circ v = \mathrm{id}$, i.e. $v(x) \in T_xX$ (physicists: *contravariant* field). In a card (U, h), v is described by the component functions $(v^1, \dots, v^n)$; under a card change $k \circ h^{-1}$ the components transform with the *differential* of the card change.

9 Pfaffian Forms & Line Integrals

Definition 9.1 (Pfaffian forms)**.** A *Pfaffian form* (1-form) on a manifold X assigns to every $x \in X$ a linear form ω_x on T_xX, i.e. $\omega : TX \to \mathbb{R}$ linear on each fibre (physicists: *covariant* field). In a card (U, h):

$$\omega = g_1\,\mathrm{d}h_1 + \cdots + g_n\,\mathrm{d}h_n, \qquad \text{row } g(z) = \big(g_1(z), \dots, g_n(z)\big).$$

Under a card change the row transforms with the inverse of the differential — precisely the opposite of vector fields.

Definition 9.2 (Exact forms)**.** For differentiable $f : X \to \mathbb{R}$, the second component of $Tf : TX \to T\mathbb{R} = \mathbb{R} \times \mathbb{R}$ is the Pfaffian form $\mathrm{d}f$, the *differential*; in a card $\mathrm{d}f = \sum_j \frac{\partial(f \circ h^{-1})}{\partial z_j}\,\mathrm{d}h_j$. Forms of this type are called *exact*. The n-tuples $(\frac{\partial}{\partial h_1}, \dots, \frac{\partial}{\partial h_n})$ and $(\mathrm{d}h_1, \dots, \mathrm{d}h_n)$ are dual bases of T_xX and $(T_xX)^*$.

Corollary 9.1 (Integrability condition)**.** *A differentiable $\omega = g_1\,\mathrm{d}x_1 + \cdots + g_n\,\mathrm{d}x_n$ can be exact only if*

$$\frac{\partial g_j}{\partial x_i} = \frac{\partial g_i}{\partial x_j} \quad \text{for all } i \neq j.$$

Example 9.1. $\omega = r\,\mathrm{d}\theta = \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2+y^2}$ on $\mathbb{R}^2 \setminus \{0\}$ satisfies the integrability condition (check in polar coordinates) but is *not* exact.

Definition 9.3 (Curve integral)**.** For a continuous Pfaffian form ω on X and a C^1-curve $\gamma : [a, b] \to X$:

$$\int_\gamma \omega \coloneqq \int_a^b \omega_{\gamma(t)}\big(\dot\gamma(t)\big)\,\mathrm{d}t.$$

Reparametrization: $\int_{\gamma\circ s} \omega = \int_{\gamma|_{[s(a),s(b)]}} \omega$ for C^1 parameter changes s.

Lemma 9.1 (Fundamental theorem for line integrals)**.** *For differentiable f and a C^1-path γ from a to b:*

$$\int_\gamma \mathrm{d}f = f(b) - f(a).$$

Hence for exact ω the integral depends only on the endpoints; in particular $\int_\gamma \omega = 0$ along closed paths. Conversely, path independence on all of X implies exactness.

Example 9.2 (Winding number)**.** For the same $\omega = r\,\mathrm{d}\theta$ and the circle $\gamma(t) = (\cos t, \sin t)$, $t \in [0, 2\pi]$:

$$\oint_\gamma \omega = \int_0^{2\pi} \mathrm{d}t = 2\pi \neq 0,$$

confirming non-exactness on $\mathbb{R}^2\setminus\{0\}$ (yet ω is locally exact by the Poincaré lemma).

10 Alternating Multilinear Forms

Definition 10.1. A *multilinear form of degree k* on V is $\omega : V^k \to \mathbb{R}$ linear in each variable; it is *alternating* if $\omega(v_1, \dots, v_k) = 0$ whenever two arguments coincide. Alternating forms form a vector space $\mathrm{Alt}^k V$; $\mathrm{Alt}^0 V = \mathbb{R}$. Examples: bilinear forms with $k = 2$; the determinant as an element of $\mathrm{Alt}^n \mathbb{R}^n$.

Proposition 10.1. *For $\omega \in \mathrm{Alt}^k V$:*

- *$\omega = 0$ on linearly dependent k-tuples; in particular $\mathrm{Alt}^k V = \{0\}$ for $k > n$;*
- *permuting arguments multiplies by the sign: $\omega(v_{\sigma(1)}, \dots, v_{\sigma(k)}) = \mathrm{sgn}(\sigma)\,\omega(v_1, \dots, v_k)$;*
- *for linear $f : \mathrm{Lin}(v_1, \dots, v_k) \to \mathrm{Lin}(v_1, \dots, v_k)$: $\omega(f(v_1), \dots, f(v_k)) = \det f \cdot \omega(v_1, \dots, v_k)$.*

Definition 10.2 (Wedge product)**.** For $\omega \in \mathrm{Alt}^k V$, $\eta \in \mathrm{Alt}^l V$:

$$\omega \wedge \eta \coloneqq \frac{1}{k!\,l!} \sum_{\sigma \in \mathrm{Sym}_{k+l}} \mathrm{sgn}(\sigma)\,(\omega \otimes \eta)\big(v_{\sigma(1)}, \dots, v_{\sigma(k+l)}\big) \in \mathrm{Alt}^{k+l} V.$$

Bilinear, associative, and *graded commutative*, that is $\omega \wedge \eta = (-1)^{kl}\,\eta \wedge \omega$.

Proposition 10.2 (Determinant formula, bases, pullback)**.**

- *Determinant formula: for linear forms $\lambda^1, \dots, \lambda^k \in \mathrm{Alt}^1 V$,*

$$(\lambda^1 \wedge \cdots \wedge \lambda^k)(v_1, \dots, v_k) = \det\big(\lambda^i(v_j)\big).$$

- *If $(\mathrm{d}h_1, \dots, \mathrm{d}h_n)$ is a basis of V^*, the products $\mathrm{d}h_{i_1} \wedge \cdots \wedge \mathrm{d}h_{i_k}$ with $i_1 < \cdots < i_k$ form a basis of $\mathrm{Alt}^k V$; in particular $\dim \mathrm{Alt}^k V = \binom{n}{k}$.*
- *For linear $f : V \to W$ the pullback obeys $f^*(\omega \wedge \eta) = f^*\omega \wedge f^*\eta$.*

Example 10.1. Cartesian coordinates on $\mathbb{R}^3$ give the bases

$$1; \quad \mathrm{d}x, \mathrm{d}y, \mathrm{d}z; \quad \mathrm{d}y\wedge\mathrm{d}z,\ \mathrm{d}z\wedge\mathrm{d}x,\ \mathrm{d}x\wedge\mathrm{d}y; \quad \mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z$$

of $\mathrm{Alt}^0, \mathrm{Alt}^1, \mathrm{Alt}^2, \mathrm{Alt}^3(\mathbb{R}^3)$ respectively ($\dim \mathrm{Alt}^2\mathbb{R}^3 = 3$, $\dim \mathrm{Alt}^3\mathbb{R}^3 = 1$).

11 Differential Forms & the Cartan Differential

Definition 11.1 (Differential forms)**.** A *k-form* on X is a map $\omega : T^kX \to \mathbb{R}$ whose restriction to each fibre is an alternating k-linear form on T_xX:

$$\omega_x(v_1,\dots,v_k) = \omega(x, v_1,\dots,v_k).$$

0-forms are functions; 1-forms are Pfaffian forms. Wedge product and pullback ($f^*\omega \coloneqq \omega\circ T^kf$) are defined fibrewise.

Remark 11.1 (Dictionary in $\mathbb{R}^3$). Under the Euclidean identification of 1-forms $\omega = a\,\mathrm{d}x + b\,\mathrm{d}y + c\,\mathrm{d}z$ and of 2-forms $\omega = a\,\mathrm{d}y\wedge\mathrm{d}z + b\,\mathrm{d}z\wedge\mathrm{d}x + c\,\mathrm{d}x\wedge\mathrm{d}y$ with vector fields (a,b,c): *contravariant* vector fields (velocities, force fields) are genuine vector fields; *covariant* fields (current densities, force fields measured on surfaces) are 1- or 2-forms. Transformation behavior under affine coordinate changes differs accordingly.

Remark 11.2 (Orientation). An orientation of T_xX (equivalently of X) is a consistent choice of "positively oriented" bases; orientation of a manifold means orienting all tangent spaces continuously. Antipodal map $x\mapsto -x$ on S^n is orientation-preserving iff n is odd.

Theorem 11.1 (Cartan differential)**.** *There is exactly one family of linear maps $\mathrm{d} : A^kX \to A^{k+1}X$ ($k\in\mathbb{N}$) with:*

- $\mathrm{d}f$ *is the differential of f (for 0-forms f);*
- *product rule:* $\mathrm{d}(\omega\wedge\eta) = \mathrm{d}\omega\wedge\eta + (-1)^k\omega\wedge\mathrm{d}\eta$ *for $\omega\in A^kX$;*
- $\mathrm{d}\circ\mathrm{d} = 0$.

It is natural: $f^*(\mathrm{d}\omega) = \mathrm{d}(f^*\omega)$ for every differentiable f.

Proposition 11.1 (Disguises in $\mathbb{R}^3$)**.** *Identifying* 1- *and* 2*-forms with vector fields (see above), the Cartan differential becomes the classical operators:*

$$\mathrm{d}^0 = \mathrm{grad}, \qquad \mathrm{d}^1 = \mathrm{rot}, \qquad \mathrm{d}^2 = \mathrm{div}\,.$$

Then $\mathrm{d}\,\mathrm{d} = 0$ *reads* $\mathrm{rot}\circ\mathrm{grad} = 0$ *and* $\mathrm{div}\circ\mathrm{rot} = 0$*: gradient fields are curl-free, curl fields are divergence-free.*

Definition 11.2 (Closed and exact forms)**.** $\omega\in A^kX$ is *closed* if $\mathrm{d}\omega = 0$; for $k>0$ it is *exact* if $\omega = \mathrm{d}\eta$ for some $\eta\in A^{k-1}X$. Every exact form is closed; the converse fails globally (e.g. $r\,\mathrm{d}\theta$).

Theorem 11.2 (Poincaré lemma)**.** *If $X\subseteq\mathbb{R}^n$ is open and* star-shaped *with respect to $a\in X$ (all segments $(1-t)a+tx$ stay in X), then every closed k-form, $k>0$, is exact.*

Sketch. *Assume $a=0$ and average along rays: the homotopy operator $H\omega = \int_0^1 t^{k-1}\omega_{tx}(x,\cdot)\,\mathrm{d}t$ (inserting x as first argument) satisfies $\mathrm{d}H + H\,\mathrm{d} = \mathrm{id}$; for closed ω, $\mathrm{d}(H\omega) = \omega$.*

Corollary 11.1. *Every closed k-form is* locally *exact: around each point there is a card in which it has a primitive (local primitives need not glue to a global one).*

12 Integration of Differential Forms

Theorem 12.1 (Substitution theorem)**.** *Let $\Phi : U\to\Phi(U)$ be a C^1-diffeomorphism between open subsets of $\mathbb{R}^n$ and $f:\Phi(U)\to\mathbb{R}$ integrable. Then $(f\circ\Phi)\cdot|\det D\Phi|$ is integrable over U, and*

$$\int_{\Phi(U)} f = \int_U (f\circ\Phi)\cdot|\det D\Phi|.$$

Corollary 12.1 (Measure transformation)**.** *For $X\subseteq U$ with $\Phi(X)$ measurable, $\mu(\Phi(X))<\infty$ iff $|\det D\Phi|$ is integrable over X; then*

$$\mu\big(\Phi(X)\big) = \int_X |\det D\Phi(x)|\,\mathrm{d}x.$$

Definition 12.1 (Integral of an n-form)**.** For $\omega = f\,\mathrm{d}x_1\wedge\cdots\wedge\mathrm{d}x_n$ on open $U\subseteq\mathbb{R}^n$ (with unique coefficient function f) and $X\subseteq U$:

$$\int_X\omega \coloneqq \int_X f\,\mathrm{d}x \quad \text{(if this exists)}.$$

Example 12.1 (Volume elements)**.** Polar coordinates: $\mathrm{d}x\wedge\mathrm{d}y = r\,\mathrm{d}r\wedge\mathrm{d}\theta$; spherical coordinates: $\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z = r^2\sin\theta\,\mathrm{d}r\wedge\mathrm{d}\theta\wedge\mathrm{d}\varphi$. This is what makes "integrating $f(r,\theta,\varphi)$ in spherical coordinates" precise.

Remark 12.1 (Stokes' theorem). For a *bordered manifold Y* (locally modelled on half-spaces; arises e.g. as $f^{-1}((-\infty,b])$ for a regular value b) with boundary ∂Y, and a C^1 $(n-1)$-form ω:

$$\int_Y \mathrm{d}\omega = \int_{\partial Y}\omega,$$

generalizing the fundamental theorem of calculus: the exterior derivative and the boundary are adjoint.

13 Higher Derivatives & Taylor

Definition 13.1 (Multiindices)**.** For $j=(j_1,\dots,j_n)\in\mathbb{N}^n$ write $X^j = X_1^{j_1}\cdots X_n^{j_n}$, $|j| = j_1+\cdots+j_n$, $j! = j_1!\cdots j_n!$. A *polynomial* of degree d in n variables is $f = \sum_{|j|\le d} a_jX^j$; e.g. the general degree-2 polynomial in three variables has the constant, linear ($a_{100}X+\dots$) and quadratic ($a_{200}X^2 + a_{011}YZ+\dots$) parts.

Lemma 13.1. *The polynomial $f = \sum_{|j|\le d} a_jX^j$ has at* 0 *the derivatives $D^kf(0) = k!\,a_k$ for all $k\in\mathbb{N}^n$.*

Definition 13.2 (Taylor polynomial)**.** For a C^k-function $f:X\to\mathbb{R}$ ($X\subseteq\mathbb{R}^n$ open) and $a\in X$:

$$T_a^kf(X) \coloneqq \sum_{|j|\le k}\frac{D^jf(a)}{j!}(X-a)^j.$$

It is the *unique* polynomial of degree $\le k$ whose partial derivatives at a agree with those of f up to order k. For C^k-maps $f:X\to\mathbb{R}^p$ apply component-wise.

Definition 13.3 (Hessian)**.** For a C^2-function f the *Hessian matrix* at a is

$$H_f(a) \coloneqq \Big(D_rD_sf(a)\Big)_{r,s}\in\mathrm{Sym}(n,\mathbb{R});$$

by the symmetry of mixed derivatives it is symmetric.

Proposition 13.1 (Calculus for Taylor polynomials)**.**

- *Chain rule: $T^k_{f(a)}(g\circ f) = T^k_{f(a)}g\circ T^k_af$ (truncate after composing).*
- *Product rule: for bilinear products, $T^k_a(f\cdot g) = T^k_a\big(T^k_af\cdot T^k_ag\big)$.*
- *Invariance: if a is a critical point and h a card around a, then $H_f(a) = Dh(a)^t\cdot H_{f\circ h^{-1}}(0)\cdot Dh(a)$: the Hessian form at a critical point does not depend on the card.*

Theorem 13.1 (Taylor's theorem)**.** *Let $f : X \to \mathbb{R}^p$ be C^{k+1} and suppose the segment from a to x lies in X. Then*

$$f(x) - T_a^k f(x) = \sum_{|j|=k+1} \frac{(x-a)^j}{j!} \int_0^1 (1-t)^k \, D^j f\big((1-t)a + tx\big) \, \mathrm{d}t.$$

There also is a Lagrange remainder*: some $t \in [0,1]$ satisfies*

$$f(x) - T_a^k f(x) = \sum_{|j|=k+1} \frac{D^j f((1-t)a + tx)}{j!} (x-a)^j.$$

Sketch. *Apply the one-dimensional Taylor formula to $\gamma(t) = f((1-t)a + tx)$ and unwind the chain rule.*

Lemma 13.2 (Divisible functions)**.** *If X is star-shaped with respect to a and $T_a^{k-1} f = 0$ ($f \in C^l$, $1 \le k \le l$), then there are C^{l-k}-functions f_j with*

$$f(x) = \sum_{|j|=k} (x-a)^j f_j(x);$$

in particular $f(x) = \sum_j (x-a)_j f_j(x)$ if all first-order derivatives vanish at a.

Example 13.1. The C^∞-function $f(x) = 0$ ($x \le 0$), e^{-1/x^2} ($x > 0$) has trivial Taylor series at 0 yet is not zero: arbitrarily many factors x can be pulled out of f. An analytic function with this property would vanish identically.

14 Morse Points

Definition 14.1 (Morse points)**.** A critical point a of the differentiable $f : X \to \mathbb{R}$ is a *Morse point* if the Hessian form $H_a f$ has maximal rank n (is nondegenerate). If all critical points of f are Morse points, f is a *Morse function.*

Theorem 14.1 (Morse lemma)**.** *If a is a Morse point of f, there is a card (U, h) around a in which f becomes a pure quadratic form:*

$$(f \circ h^{-1})(x) = f(a) - x_1^2 - \cdots - x_\sigma^2 + x_{\sigma+1}^2 + \cdots + x_n^2.$$

The number $\sigma \in \{0, \dots, n\}$, the number of negative squares (= negative eigenvalues of the Hessian), is the Morse index *of a. Near a regular point f is linear in a suitable card; near a Morse point it is quadratic.*

Proposition 14.1. *Every Morse point is an* isolated *critical point: some neighborhood of a contains no further critical point of f.*

Theorem 14.2 (Extremum criteria)**.**

- *If f has a local extremum at a, then a is a critical point ($Df(a) = 0$).*
- *If a is a Morse point, then f has a local minimum at a iff the Morse index is 0, and a local maximum iff the index is n.*

Lemma 14.1. *If the Hessian form $H_a f$ has at least one negative eigenvalue, then f has no local minimum at a (dually for maxima).*

Sketch. *Write $f(x) = \frac{1}{2} x^t H(x) x$ with continuous H; along an eigenvector for a negative eigenvalue f takes negative values arbitrarily close to a.*

Example 14.1 (Quadratic forms on the sphere)**.** Critical points of $q|_{S^{n-1}}$ are the unit eigenvectors, with the eigenvalues as critical values; the smallest and largest eigenvalues give the global min and max. Such a critical point is Morse iff the eigenvalue is simple. Check definiteness of $H_a f$ with the Sylvester criterion.

Remark 14.1 (Recipe for global extrema). Find the critical points (with Lagrange multipliers for constraints), test the Hessian for index — and treat boundary/constraint parts *separately*: the theorems apply only on the (borderless) manifold. For $f(x,y) = x^2 + x + 2y^2$ on the closed disk: interior critical point $(-\frac{1}{2}, 0)$ with local min $-\frac{1}{4}$; boundary extrema via a Lagrange multiplier on S^1.

15 Quick Reference: Coordinates & Vector Analysis

- Polar: $x = r\cos\theta$, $y = r\sin\theta$, $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$; cylindrical: add z, $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z$; spherical: $(r\sin\theta\cos\varphi,\ r\sin\theta\sin\varphi,\ r\cos\theta)$, volume element $r^2 \sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi$.
- Dictionary $\mathbb{R}^3$: functions = 0-forms, d^0 = grad; vector fields $\leftrightarrow$ 1-forms, d^1 = rot; vector fields $\leftrightarrow$ 2-forms, d^2 = div; rot grad = 0, div rot = 0; a field with scalar potential is curl-free, one with vector potential is divergence-free.
- Closed vs. exact: closed $\neq$ exact globally ($r\,\mathrm{d}\theta$ on $\mathbb{R}^2 \setminus \{0\}$, $\oint = 2\pi$); on star-shaped domains closed = exact (Poincaré); local exactness always holds.
- Convergence theorems: monotone (bounded integrals) and dominated ($|f_j| \le g \in L^1$) convergence; exhaustion for unbounded domains; improper integrals require *absolute* limits.
- Key tools: Newton–Leibniz, Fubini (only for integrable f!), substitution/Jacobi determinant, Cavalieri, Stokes $\int_Y \mathrm{d}\omega = \int_{\partial Y} \omega$, Taylor with integral or Lagrange remainder, Morse lemma.
- Topology: compact = closed + bounded; continuous images of compact/connected sets are compact/connected; on compact sets continuous real functions attain min and max; continuous bijections out of compact spaces are homeomorphisms.